Tab 1

```
#include<stdio.h>

int main()
{

int distance, hour, avg;
printf("enter the distance");

scanf("%d", &distance);

printf( " enter the hour");
scanf("%d", &hour);


avg = distance/hour;

printf(" speed of car %d", avg);


return 0;

}
```

Main.c

```c
#include<stdio.h>

int main()
{

int distance, hour, avg;
printf("enter the distance");

scanf("%d", &distance);

printf( " enter the hour");
scanf("%d", &hour);


avg = distance/hour;

printf(" speed of car %d", avg);


return 0;

}
```

Console I/O AI Agent

```
enter the distance 20
 enter the hour 1
 speed of car 20
```

tab 2

```c
#include <stdio.h>

int main()
```

```c
{
 int num1, num2, temp;

 printf("Enter value of num1: ");
scanf("%d", &num1);

 printf("Enter value of num2: ");
scanf("%d", &num2);

 // Swap the values
 temp = num1;
 num1 = num2;
 num2 = temp;

 printf("\nAfter swapping");
printf("num1 = %d\n", num1);
printf("num2 = %d\n", num2);

 return 0;
}
```

```c
#include <stdio.h>

int main()
{
    int num1, num2, temp;

    printf("Enter value of num1: ");
    scanf("%d", &num1);

    printf("Enter value of num2: ");
    scanf("%d", &num2);

    // Swap the values
    temp = num1;
    num1 = num2;
    num2 = temp;

    printf("\nAfter swapping");
    printf("num1 = %d\n", num1);
    printf("num2 = %d\n", num2);

    return 0;
}
```

```
Enter value of num1: 10
Enter value of num2: 20

After swappingnum1 = 20
num2 = 10
```

tab 3

```c
#include<stdio.h>

int main()
{
float taxrate , salary;
float taxAmount , netIncome;
```

```
printf ("enter the salary ");
scanf("%f", &salary);

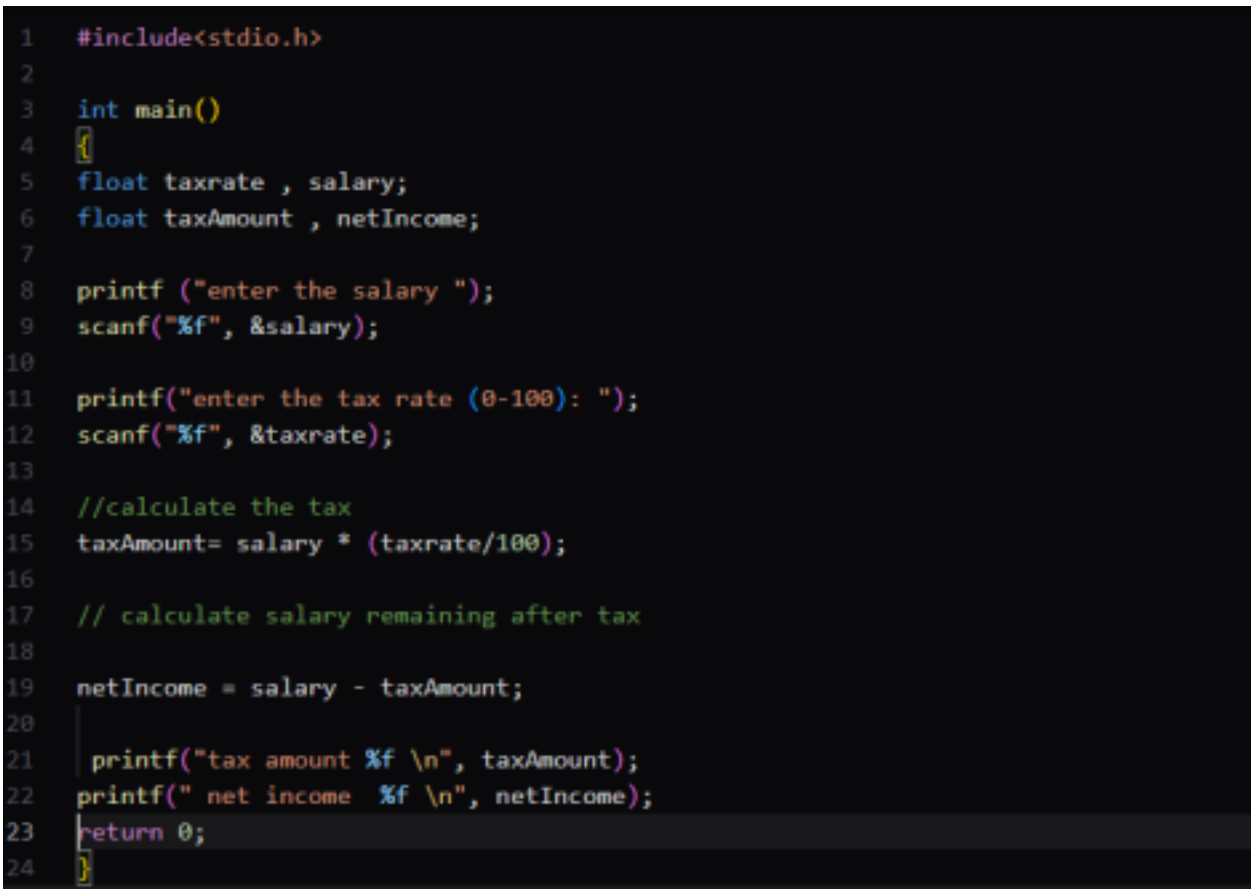
printf("enter the tax rate (0-100): ");
scanf("%f", &taxrate);

//calculate the tax
taxAmount= salary * (taxrate/100); //

calculate salary remaining after tax

netIncome = salary - taxAmount;

 printf("tax amount %f \n", taxAmount);
printf(" net income %f \n", netIncome);
return 0;
}
```



```c
#include<stdio.h>

int main()
{
float taxrate , salary;
float taxAmount , netIncome;

printf ("enter the salary ");
scanf("%f", &salary);

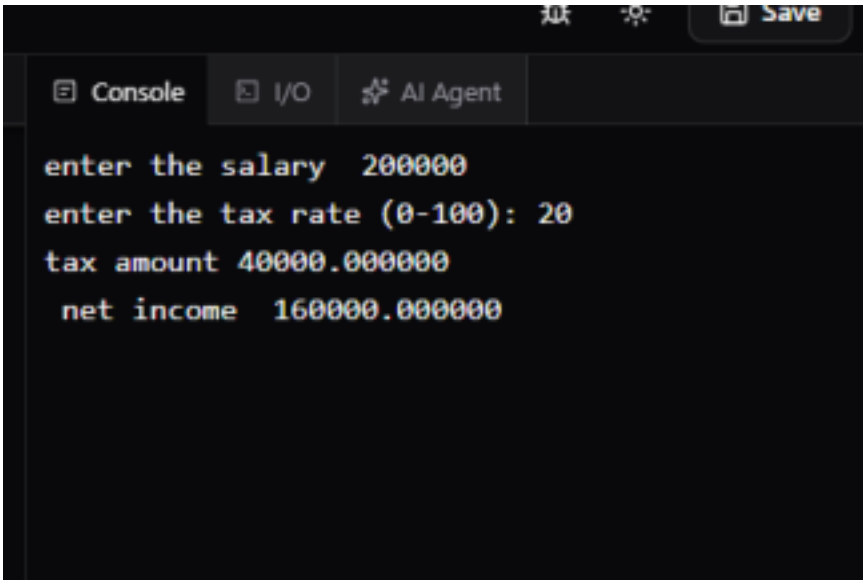
printf("enter the tax rate (0-100): ");
scanf("%f", &taxrate);

//calculate the tax
taxAmount= salary * (taxrate/100);

// calculate salary remaining after tax

netIncome = salary - taxAmount;

 printf("tax amount %f \n", taxAmount);
printf(" net income  %f \n", netIncome);
return 0;
}
```



tab 4

```c
#include <stdio.h>

int main()
 {
 int pennies = 368;
 int quarters, dimes, nickels, remaining;

 // Calculate quarters
 quarters = pennies / 25;
 remaining = pennies % 25;

 // Calculate dimes
 dimes = remaining / 10;
 remaining = remaining % 10;

 // Calculate nickels
 nickels = remaining / 5;
 remaining = remaining % 5;

 // Remaining pennies
 pennies = remaining;

 printf("Quarters: %d\n", quarters);
printf("Dimes: %d\n", dimes);
printf("Nickels: %d\n", nickels);
```

```
printf("Pennies: %d\n", pennies);

 return 0;
}
```

```c
int main()
 {
    int pennies = 368;
    int quarters, dimes, nickels, remaining;

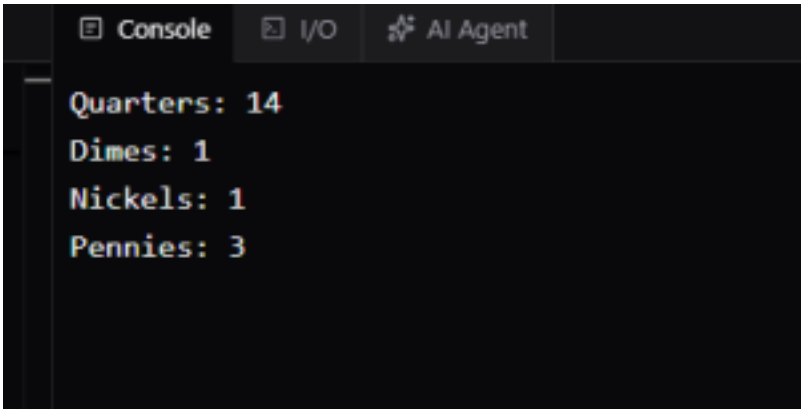
    // Calculate quarters
    quarters = pennies / 25;
    remaining = pennies % 25;

    // Calculate dimes
    dimes = remaining / 10;
    remaining = remaining % 10;

    // Calculate nickels
    nickels = remaining / 5;
    remaining = remaining % 5;

    // Remaining pennies
    pennies = remaining;

    printf("Quarters: %d\n", quarters);
    printf("Dimes: %d\n", dimes);
    printf("Nickels: %d\n", nickels);
    printf("Pennies: %d\n", pennies);
```



tab 5..

```c
#include <stdio.h>
#include <math.h>
```

```c
int main() {
 double a, b, c;
 double discriminant;
 double x1, x2;

 printf("Enter values of a, b, and c: ");
scanf("%lf %lf %lf", &a, &b, &c);

 // Calculate the discriminant
 discriminant = b * b - 4 * a * c;

 // Calculate both roots
 x1 = (-b + sqrt(discriminant)) / (2 * a);
x2 = (-b - sqrt(discriminant)) / (2 * a);

 printf("Root x1 = %.2f\n", x1);
printf("Root x2 = %.2f\n", x2);

 return 0;
}
```

```c
#include <stdio.h>
#include <math.h>

int main() {
    double a, b, c;
    double discriminant;
    double x1, x2;

    printf("Enter values of a, b, and c: ");
    scanf("%lf %lf %lf", &a, &b, &c);

    // Calculate the discriminant
    discriminant = b * b - 4 * a * c;

    // Calculate both roots
    x1 = (-b + sqrt(discriminant)) / (2 * a);
    x2 = (-b - sqrt(discriminant)) / (2 * a);

    printf("Root x1 = %.2f\n", x1);
    printf("Root x2 = %.2f\n", x2);

    return 0;
}
```

Console | I/O | AI Agent

```
Enter values of a, b, and c: 2 3 4
Root x1 = nan
Root x2 = nan
```

tab6

```c
#include <stdio.h>
int main()
{
 int a, b, c;
 float result_a , result_b, result_c;

 printf(" enter the first integer");
scanf("%d", &a);

 printf("enter the second integer");
scanf("%d", &b);
```

```c
 result_a = a/b;
 printf( "a/b = %0.2f\n", result_a);

 result_b = (float) a/b;
 printf(" (float) a/b = %.3f\n", result_b);

 result_c= (float) (a/b);
 printf(" (float) (a/b) = %0.2f\n", result_c);

return 0;

}
```

Main.c

```c
#include <stdio.h>
int main()
{
    int a, b, c;
    float result_a , result_b, result_c;

    printf(" enter the first integer");
    scanf("%d", &a);

    printf("enter the second integer");
    scanf("%d", &b);

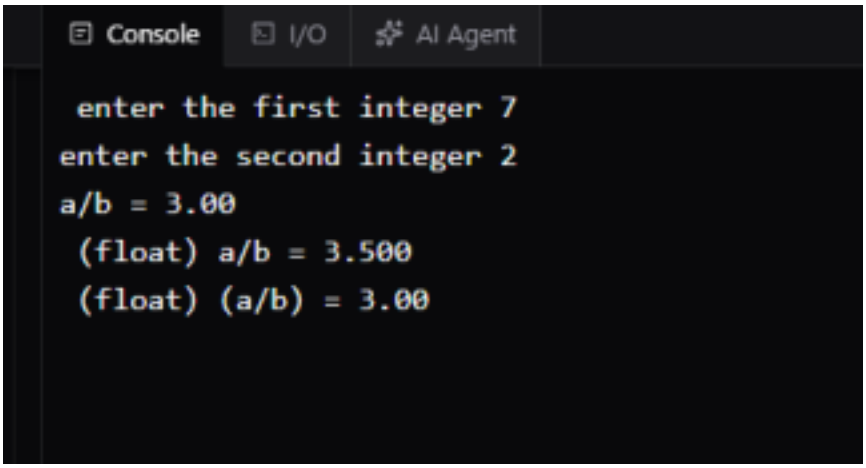
    result_a = a/b;
    printf(  "a/b = %0.2f\n", result_a);

    result_b = (float) a/b;
    printf(" (float) a/b = %.3f\n", result_b);

    result_c= (float) (a/b);
    printf(" (float) (a/b) = %0.2f\n", result_c);

    return 0;

}
```



Tab 7

```c
#include<stdio.h>

int main ()
{
 int mark1, mark2, mark3;
 int total;
 float average;

printf(" enter the marks of three subjects");
scanf(" %d\n, %d\n,%d\n", &mark1, &mark2, mark3);

// total marks
total = mark1 + mark2 + mark3 ;

printf("total marks is %d\n", total);


// avaerage of marks

average = (float) total/3;

printf("average marks is %f\n", average);

return 0;
}
```

Complier does not execute its code

```c
#include<stdio.h>

int main ()
{
  int mark1, mark2, mark3;
  int total;
  float average;


printf(" enter the marks of three subjects");
scanf(" %d\n, %d\n,%d\n", &mark1, &mark2, mark3);

// total marks
total = mark1 + mark2 + mark3 ;
printf("total marks is %d\n", total);

// avaerage of marks

average = (float) total/3;
printf("average marks is %f\n", average);


return 0;
}
```

Tab 8

```c
#include <stdio.h>

int main()
{
 float value;
 int converted;
 float difference;

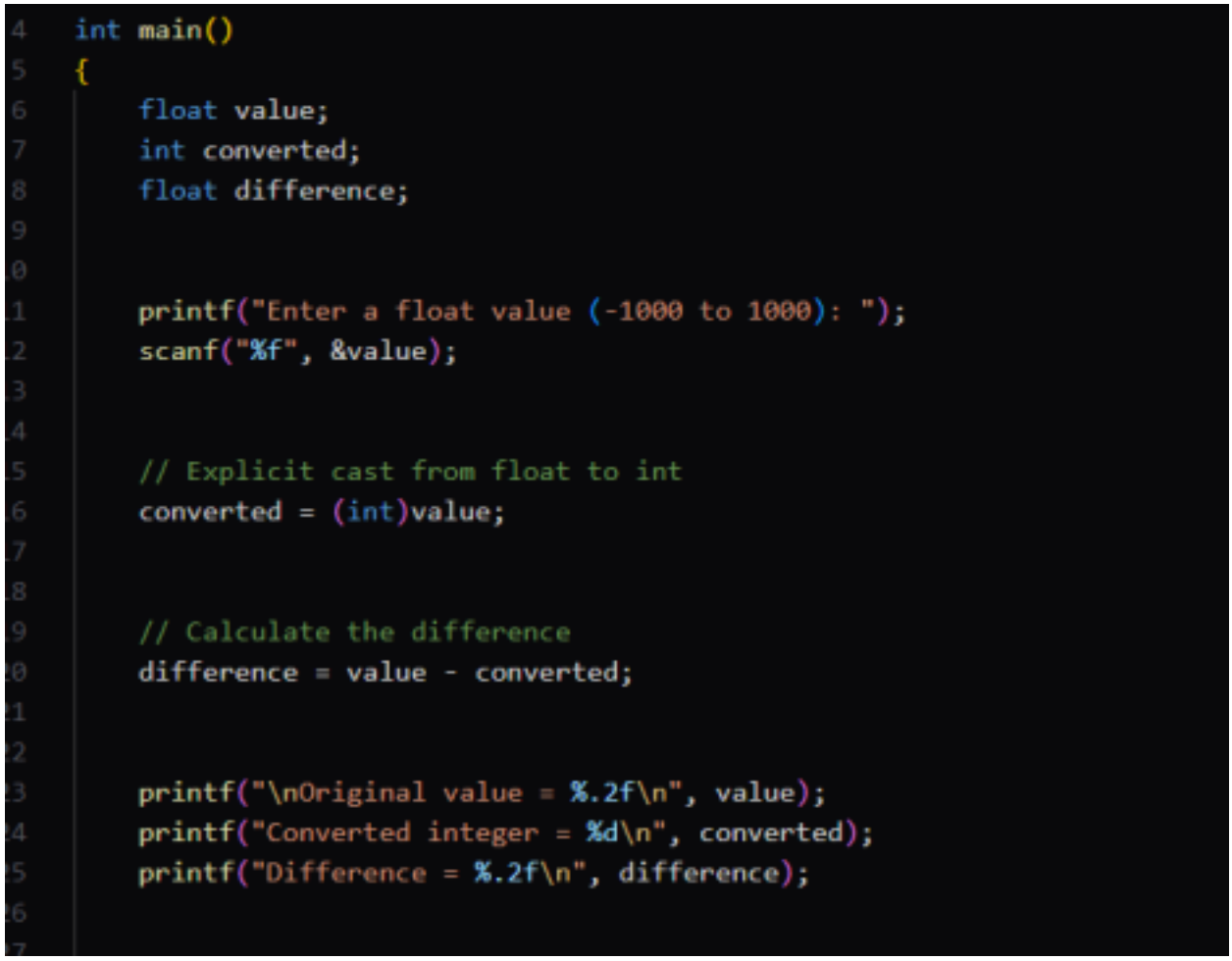
 printf("Enter a float value (-1000 to 1000): ");
scanf("%f", &value);
```

```c
 // Explicit cast from float to int
 converted = (int)value;

 // Calculate the difference
 difference = value - converted;

 printf("\nOriginal value = %.2f\n", value);
printf("Converted integer = %d\n", converted);
printf("Difference = %.2f\n", difference);

 return 0;
}
```



```
Enter a float value (-1000 to 1000): 400 500

Original value = 400.00
Converted integer = 400
Difference = 0.00
```